

Item Code Studio

Sprint Retrospective · 6–7 August 2026 · Agent 0 (Foundation) + Agents A–H, run in parallel

Companion to **Sprint_Plan_v3.xlsx** (now carries this same retro, added per-story, in five new columns on the Delivery Plan tab) and to each agent's own handover in **agents/done/AGENT_*.md**. This document is the narrative version: what happened, in what order, what worked, what didn't, and what changed from the plan along the way.

1. Executive summary

	Stories	Points
Built and verified this session	52	240
Pre-existing capability, touched/re-verified this session	22	104
Partially built (real, tested groundwork; not fully wired)	2	13
Still blocked on Anuraag (correctly not attempted)	19	87
Gap — in scope, flagged, deliberately not built	2	7
Deferred (backlog, correctly out of scope)	2	8
Total tracked	99	459

Nine work packets (Foundation + A–H) ran across roughly nine hours of elapsed session time, spanning two calendar days, two systemic interruptions, and one security incident that was caught and fixed inside the same session. Every packet reached a written handover (**agents/done/AGENT_*.md**); the final integration run (86/86 smoke tests, a live login→issue→edit→revert round trip, tier failover proven with real processes) passed clean.

The single most consequential open item: `core/resolve.py`'s ERPNext push path currently sends our own 889-group taxonomy name as ERPNext's `item_group`. ERPNext only has ~23 groups. Every real push will be correctly refused until an operator-facing sub-head→Item-Group mapping exists on the create screen. Agent G found this, declined to invent the mapping itself (per **CONTRACTS.md** rule 4 — never push our taxonomy), and flagged it plainly. Nothing is broken; nothing has silently mismapped anything; it simply isn't wired yet. See §6.

2. How “time taken” is reported, and why not more precisely

Sub-story timing was never tracked — each agent worked its whole packet as one continuous task, not story-by-story with checkpoints, so a story-level duration would be invented, not measured. What follows is honestly measurable instead:

- **Tool calls and tokens per completed segment** — a real, granular measure of effort, reported per agent in §4.
- **Number of resumes** — how many times a packet had to be restarted after an interruption. This is the most informative “friction” number in this retro: **every one of the eight parallel agents needed at least one resume**, and four needed two or three.
- **Wall-clock span** is reported at the session level (§3), not per-agent, because it is dominated by two interruptions that had nothing to do with any agent's actual work — see §5.

3. Timeline

- **Foundation (Agent 0), ~45 min, uninterrupted.** Schema migration (six new tables), the `server.py` → `routes/*.py` split, `settings/ledger config`, `theme file`. Landed clean; `agents/done/AGENT_0.md` written.
- **All eight agents (A–H) launched simultaneously**, per the original plan — genuinely parallel, since `CONTRACTS.md`'s file ownership is disjoint by design.
- **All eight stalled within ~600 seconds, producing zero file output.** A systemic failure — very likely eight concurrent heavy agents exceeding some shared capacity limit, not a fault in any individual brief. Every file on disk was still exactly what Agent 0 had left.
- **Recovery: rebatched into groups of 3.** Resumed A, B, C first; once they were producing real, healthy output, brought back D, E, F; then G and H. No further systemic stalls.
- **Two resumed agents (A, C) hit a second, unrelated failure** (“connection closed mid-response”) right after landing real progress — a transient API disconnect, not a stall. Both resumed and completed normally.
- **Security incident, caught mid-session:** Agent H's `HANDOVER.md` included a real, live admin bootstrap password in plaintext. Caught immediately on review, the password was rotated and invalidated, the document was redacted and rewritten, and the repo was swept for any other copy before anything was pushed anywhere. See §5.2.
- **Hard stop: account-level monthly spend limit hit**, killing four agents simultaneously (E, G, H, and a resumed B) mid-task. Not a transient error — work paused rather than burning further attempts against it.
- **Resumed once the limit was raised.** B finished its fix and verified it with a real Chrome round-trip; G and H finished their remaining work and each wrote a full handover. E's work was already complete on disk before its connection dropped — verified, not re-run.
- **Final integration (Agent H):** `clean seed.py --rebuild` → server starts clean → 86/86 smoke tests → 401 sweep confirmed → live login→issue→edit→revert → group-move queue-claim proven → ERPNext confirmed off → tier failover/reconciliation proven with two real server processes → `HANDOVER.md` written.
- **Retrospective + GitHub.** This document, the per-story columns in `Sprint_Plan_v3.xlsx`, and a private GitHub repository (Saisam13/item-code-studio) with real business data excluded via `.gitignore` and a pre-push secret sweep.

4. Per-agent summary

Agent	Built	Pts	Resumes	Headline deliverable
0 Foundation	5	26	0	Schema, route split, settings/ledger config, theme
A Public	5	20	2	Split landing page, decoder, directory, dictionary
B Auth	8	29	2	Login/sessions, Settings+API key screen, closed the X-User hole
C Matching	6+5 ext.	~26	2	LLM-first inversion — rules shortlist + veto, deterministic fallback
D Engine	5+6 ext.	~29	1	Queue-claim numbering, preview_code, concurrency proof, leases
E Create	6+6 pre	~25	1	Three-phase create screen, cascades, idempotent commit
F Master	9+3 pre	~50	1	Editable master, versions, revert-respects-frozen-codes
G ERPNext	6+3 pre	~34	2	Closed-list guardrail, drift sync, failsafe reconciliation
H Deploy	8+2 part.	~34	2	86/86 smoke suite, tier failover, installer, verified backup/restore

Points above are approximate per-agent sums from the story table; a few stories split across two agents' files (e.g. K11/C4) are counted once.

5. The two systemic incidents, in more detail

5.1 The 600-second stall

Launching eight heavy concurrent agents at once produced zero output from any of them after ten minutes — not a slow start, a complete stall, uniformly. Rebatching into groups of three resolved it immediately with no further systemic failures for the rest of the session. **Lesson for next time:** default to smaller concurrent batches (3–4) for heavy multi-tool agent work in this environment, rather than assuming maximum parallelism is free.

5.2 The exposed credential

Agent H's HANDOVER.md — a document explicitly meant to be read and kept — included the real first-run admin password in plaintext, with a comment telling the reader to change it “since it has been displayed in this document.” That reasoning is backwards: a document is exactly where a live credential should never go, precisely because unlike a console it persists.

What was done, in order: the password was rotated immediately (`manage.py resetpw admin`, old one invalidated); the document was rewritten to explain the mistake and point at generating a fresh credential yourself rather than repeat one in text; the whole repository was swept for any other copy before anything was pushed to GitHub. No external exposure occurred — this was caught before the document left the local machine.

Lesson for next time: an explicit instruction to any agent writing a setup/handover document should state, up front, that a live credential is handed to a person once, on a console, never written into a persistent file.

5.3 The spend-limit hard stop

A hard account-level cap, not a rate limit — four agents failed simultaneously mid-task with an identical, unambiguous error. The correct response was to stop immediately rather than retry, and wait for the limit to be raised before resuming. Work resumed cleanly from each agent's own transcript once it was — nothing had to be redone, since most agents' actual file writes had already landed on disk before the connection died.

6. What worked

- **Disjoint file ownership genuinely enabled real parallelism.** Once past the two interruptions above, nine agents editing one codebase concurrently produced no file-content conflicts — the `CONTRACTS.md` design held.
- **Agents caught each other's real bugs, unprompted** — the strongest signal that the handover discipline worked: Agent F found the Secure-cookie-over-HTTP bug in Agent B's code while browser-testing its own screen; Agent D's own concurrency test caught a real race in `mint_from_lease`; Agent F's revert testing caught an over-claimed-revert bug (`updated_at` miscounted as a “restored field”); Agent G caught that its own new guardrail code would have crashed two other agents' live call sites and fixed itself to match the existing contract; Agent B found a `D.log()` commit gap in Agent O's `core/db.py` that Agent O then fixed at the source for every caller at once.
- **Verification was real, not asserted.** Nearly every handover cites a specific command, curl call, or browser interaction. The frozen-code revert case, the lease concurrency race, and tier failover were each proven with an actual running process.
- **The deliberate design reversals** (queue-claim numbering, LLM-first matching) landed coherently across three separate files owned by three different agents with no drift between them — each independently cross-referenced `CONTRACTS.md` rather than the older `Sprint_Plan_v3.xlsx` wording, which in places still describes the pre-reversal design.

7. Challenges & deviations from the plan

- **`Sprint_Plan_v3.xlsx` itself is stale relative to agents/`CONTRACTS.md`** on at least two rows. C4/K11 describe the old “vacated number held for a $\geq 88\%$ name match” rule as the target; `CONTRACTS.md` decision #5, dated the same day, explicitly reverses this to queue-claim/lowest-first. H13 is marked NEEDS INPUT/Blocked in the plan, but the actual brief instructed building it anyway, gated behind a default-off setting — which is what happened.
- **`CONTRACTS.md` left several routes' ownership genuinely ambiguous** (`/api/group/move*`, `/api/groups`, `/settings/*` vs `/api/v1/settings`). Agent O made judgement calls and documented every one explicitly so later agents could override them; none did, but each said so plainly.
- **Several agents needed to touch a file outside their own ownership** to keep the system working — Agent B editing `core/dispatch.py` to close the X-User hole everywhere at once, Agent D fixing call sites after changing a frozen function's signature. Flagged explicitly in every case rather than done silently.
- **The ERPNext item-group mapping gap (§1) is a real, unresolved deviation** from a working end-to-end system, not a bug — the one place where “correctly refuses to guess” and “not yet buildable by anyone in this session” collide.

8. Open items — for Anuraag

- **ERPNext item_group mapping** — needed before any real ERPNext push succeeds. Needs an operator-facing field on the create screen.
- **seed.py non-determinism** — identical reruns of `seed.py --rebuild` can land on either 889/1,947 or 891/1,949 groups/items. Not caused by this session; looks like Python's per-run hash randomization affecting a set/dict-ordered dedup step.
- **core/tier.py's continuous wiring into server.py**, and the tier-to-tier shared secret, are both built and tested standalone but not yet connected — a documented small patch to a file only Agent O owns.

- **Legacy un-versioned mutating routes remain unauthenticated** — deliberate, so the pre-existing UI didn't break mid-build, but a real exposure the moment this runs on a public VPS.
- **The Item Code Specification slot-field name is an unconfirmed guess** — check against live ERPNext doctype metadata before `erp.populate_specs` is ever turned on.
- **No VPS exists yet** — `install/vps/` is written and vendor-neutral, not run against a real host.
- Everything already listed as blocked in `Sprint_Plan_v3.xlsx`'s NEEDS INPUT rows (19 stories, 87 points) is unchanged and correctly untouched.

9. Where everything is

- Per-story retro: `Sprint_Plan_v3.xlsx` → Delivery Plan tab, five new columns after Notes.
- Full technical handover: `HANDOVER.md` (restart/restore/adduser/rotate-key runbook).
- Each agent's own account: `agents/done/AGENT_0.md` through `AGENT_H.md`.
- Source, private: `github.com/Saisam13/item-code-studio` (main).
- Your new admin password was sent to you directly in chat when it was rotated — not repeated here, per the lesson in §5.2.